

Charles Eubanks

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Education

Aurora University - 4.0 GPA

Bachelor of Science in Computer Science and Mathematics

May 2026

Waubensee Community College - 4.0 GPA

Associate of Science in Computer Science

May 2023

Projects

Dungeons and Dragons Performance Assistant - Project Lead, Software Engineering Capstone

Python, FastAPI, PyTorch, React, TypeScript, MongoDB Atlas, Docker

- Led a 3-person development team by assigning feature ownership, coordinating progress against faculty-defined requirements, and reviewing teammate contributions for quality and production readiness.
- Architected and implemented core backend workflows for a full-stack combat simulation and recommendation platform in compliance with the D&D 5th edition ruleset, integrating RESTful FastAPI services, a React/TypeScript UI, and MongoDB encounter-state persistence.
- Developed a multi-criteria recommendation engine using Pareto front filtering and TOPSIS scoring across metrics such as probability of success, expected damage, and action impact ratings based on encounter-state context.
- Validated recommendation effectiveness over 160+ observed action outcomes using ANOVA and MLR, finding that top-ranked recommendations achieved 44.7% higher mean realized action outcomes than lower-ranked alternatives, with statistically significant differences across action ranking groups.
- Created repeatable test cases to validate backend behavior and support automated testing, CI/CD configuration, and deployment automation.

Dungeons and Dragons Real-time Analytics Generator - Mathematics Capstone

- Developed a Python-based D&D combat analytics engine that models action outcomes from encounter, player, and enemy parameters, calculating probability of success, expected damage, and action-impact metrics for real-time decision support.

NHL Shot Count Prediction Model

- Partnered with one teammate to compile and refine 14 years of NHL data in Python, applying machine-learning regression models with scikit-learn to forecast team shot counts with 90% accuracy.

Web-scraped Data Analysis on Smash Ultimate Characters

- Used BeautifulSoup to parse through multiple websites and pulled together data based on average and professional win-rates based on the data of all 89 characters, matchups against other characters, and the multiple stages they can play on. Performed multiple probability distributions to find a given player's chance of winning.

Battleship

- Used Object Oriented Design in JavaScript to create an iconic board-game playable via browser against bots.

Typeracer statistical analysis

- Pulled data from a web-based typeracer database using Selenium to perform statistical analysis on leaderboard data in Java, all stored inside of a cloud-based phpMyAdmin database that can be queried later for desired information.

Technical Skills

Programming Languages: Python, TypeScript, JavaScript, Java, SQL, C++, HTML/CSS

Libraries & Frameworks: React, FastAPI, PyTorch, scikit-learn, Pytest, JUnit 5, Selenium, Pandas, NumPy, Next.js, Bootstrap

Developer Tools: Git/GitHub, Docker, MongoDB Atlas, phpMyAdmin

Analytical Methods: MLR, ANOVA, Machine Learning, Excel Analysis ToolPak, Solver Optimization, Time Series, Neural Networks, Probability Distributions, TOPSIS, Pareto Optimization

Experience

Aurora University, Aurora, IL

August 2024 - May 2026

Academic Support Center Peer Tutor

- Helped students reinforce understanding of mathematical and programming materials such as tutoring calculus, object oriented design, coding techniques in various programming languages, and the testing and debugging of code in a multitude of applications.